

INFO-I300 HCI/d

Project 3

WMCS (Weight Machine Cross Synchronization)

Weight Machine Set Identifier through Near Field Communication Technology

Chae Woon Kim

RATIONALE AND RESEARCH

A great start to a day could be down to a myriad of factors, but to some, a great morning workout could make or break their day. According to a Harvard Business Review article on exercise, working out has shown to elevate mood and help prolong mental stamina (Source 16). From 2000 to 2017 over 60 million people in the US were signed up for gym memberships in the USA (Source 10). According to Statista, in 2017, 70% or more attended their gym more than twice per week and spent roughly 20 minutes or more working out (Source 20, 21). With gym memberships becoming more prevalent throughout the USA, the longer the cue individuals have to wait for their machine to open up seems a more likely occurrence as the days goes past.

Picture this scenario, you're at the gym with your pinpoint accurate routine ready to go, unaware of the fact that all the machines you need are being used by others with their own agendas. You're trying to salvage your workout by using a machine that awkwardly fits into your workout. The makeshift plan doesn't cut it and you eventually give up, you pack your belongings and set forth on an awful start to your day. The next time you're back with the hopes that things will be different, you're faced with an old friend in déjà vu.

THE PROBLEM

It's one thing to have a workout plan, but it's another to have the machines to execute your plan. With certain periods of the day being the likely destination for gym goers. The question is not about creating a workout catered to you, but whether you're able to fit in a workout when everyone is there. During the survey and interviews I conducted at a university gym, the biggest problem seems to be the lack of available machines (Source 1, 2, 3, 4).The availability of the machine is affecting gym-goers to maximize their time, in fact, people are only utilizing a third to half their time while they're occupying the equipment (Source 15).

WHAT'S AVAILABLE TODAY?

Currently there are heat maps in certain gyms to indicate the availability of cardio equipment, but this technology only visualizes the traffic in specific areas. A heatmap provides the data visualization as a way for the gym owners to understand what locations the exerciser gravitates towards, but from a user standpoint of view it doesn't offer much functionality in regards to what they want to know, which is how long it'll take for the person using the machine to get off (Source 9, 19).

Equipment developers have gone as far as to include in device tracking features that measure the usage of the particular equipment as well as built-in alert system to notify the staff when the machine is in need of maintenance (Source 14, 15).

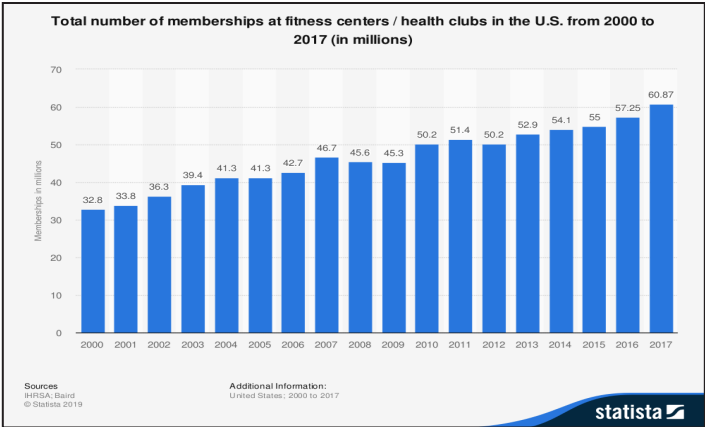


Image 8: Bar Graph of Total Gym Membership from 2012 to 2017

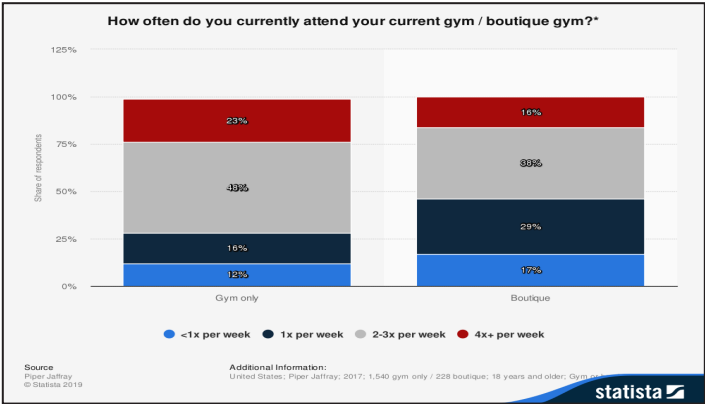


Image 7: Graph of Attendance of Current Gym

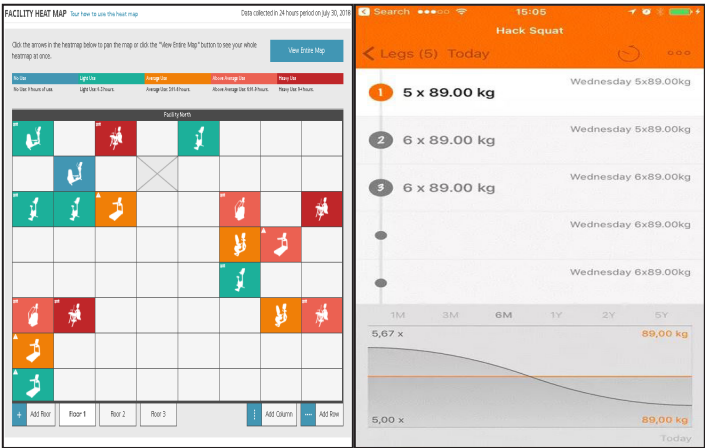


Image 10: Heat Map

Image 5: Rep Count App

TECHNOLOGY UTILIZED: NFC (NEAR FIELD COMMUNICATION)

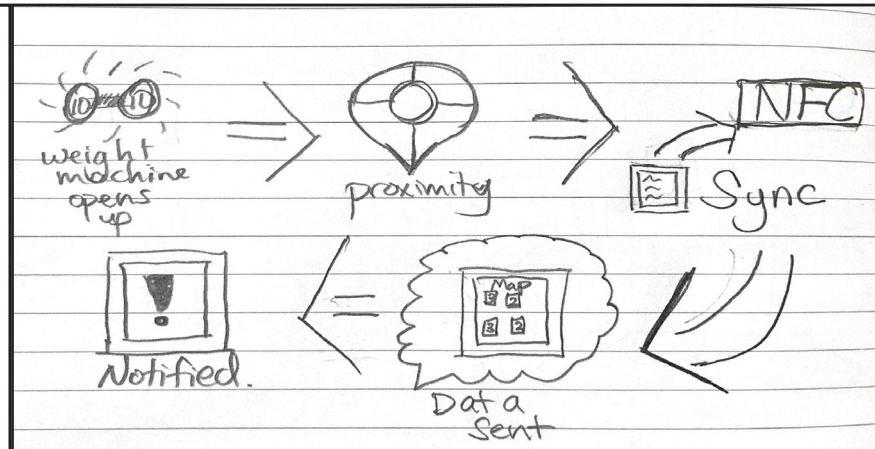
As of 2018 there were more than 1.9 billion NFC enabled mobile devices (Source 12, Image 6). According to Cnet, over 400 million cellphones were being shipped with NFC capabilities, which allows devices within a close proximity to transfer small amounts of data to one another (Source 8, 22). The mainstream use of NFC has been mostly applied in payment systems as a way to digitize wallets. However, despite the majority of the use coming in payment functionalities, NFC is not limited to payments (Source 8).

THE PROPOSITION

The idea is to add an NFC chip connected to a set counter inside a workout machine. The two devices (the user's phone and the machine) will work in tandem to provide an experience where if the gym-goer gets to a close enough proximity to the equipment, the sensor detects the phone and initiates a set counter. This detection will ping the message across an already existing gym app, which this feature will be added on top of. The map on the gym application will indicate how many sets the individual has left on the machine, allowing other users to be aware and theoretically lessen bystanders. The set counter can be toggled to the number the user wants (this function can be toggled and adjusted). The following diagram illustrates how the technology would be applied in a real-life scenario (Image 1).

The User/Phone

- The user sits down and gets into a close enough proximity to the machine where it is detected.
- The user's phone is synced with the machine via NFC.
- The user starts using the machine and the set counter initiates.



The Machine/Set Counter

- The set counter can be toggled to the users liking (all adjustable on the app).
- To prevent users from taking too much time on one machine, there is a limit to the counter.
- The data is pinged to the app where other users are aware of the availability of the machine.

Image 1: Storyboard of WMCS Technology Implementation

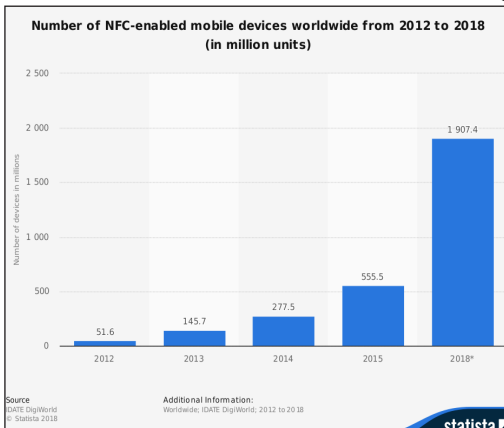


Image 6: Graph of NFC Enabled Devices

The Way the WMCS Works (Linking it All Together)

- Once the machine and phone are synchronized, the app notifies the other users (on the map screen) that the machine is taken by the user and has a certain amount of sets left.
- This will prevent bystanders and allow gym members to know if there is a surge in traffic for a particular machine.
- This is meant to be a feature that is added on top of an already existing gym application.



Image 4: NFC

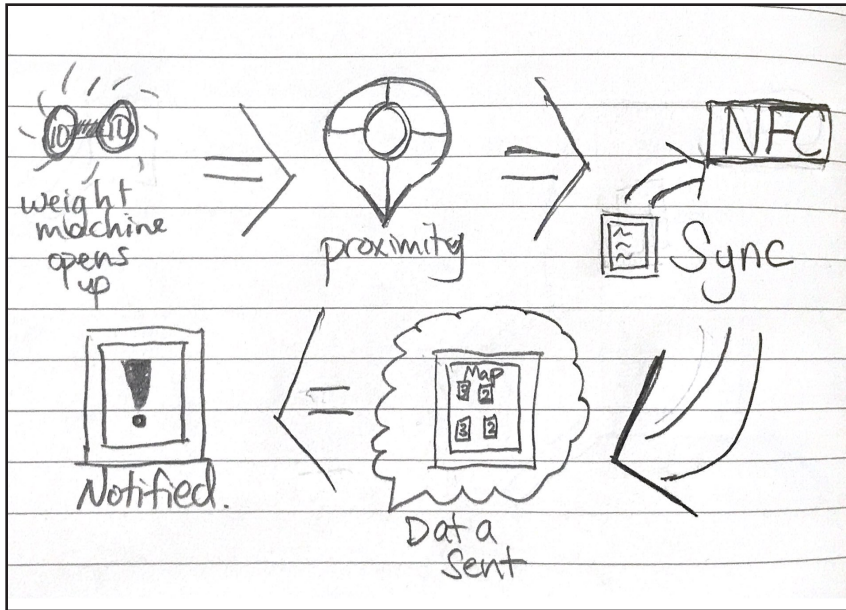


Image 1: Storyboard of WMCS Technology Implementation

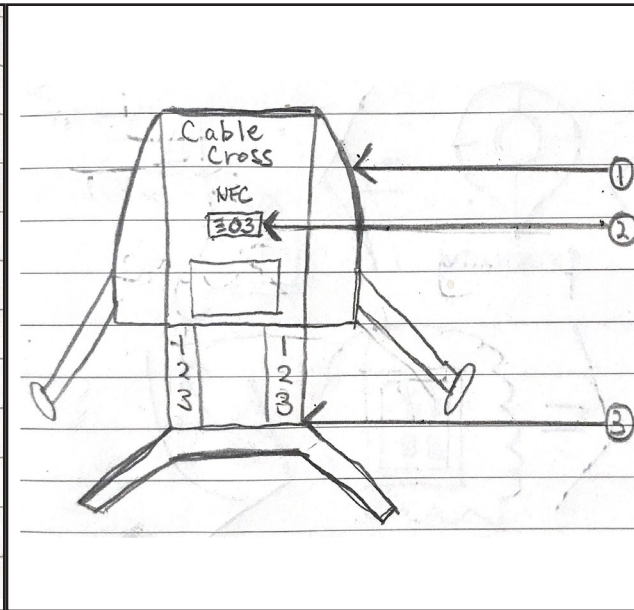


Image 3: Cable Cross Machine with WMCS

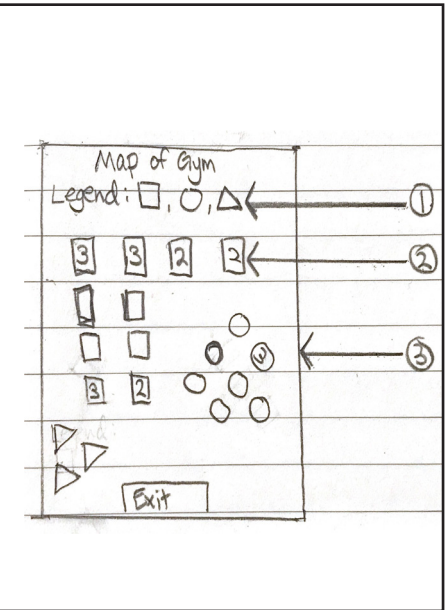


Image 2: Gym App Interface with WMCS

Image 1 Illustrates the storyboard of how the WMCS would be applied in a real-life scenario:

1. Before everything happens, the user goes to the gym.
2. The weight machine opens up for the gym-goer to use.
3. As the gym-goer reaches within the proximity of the machine, the phone is detected by the machine through NFC.
4. The WMCS technology commences and the phone is synced to the machine.
5. Data is sent to the gym application (that the WMCS is implemented on) and the set counter is initiated.
6. Other users of the gym app are notified.

Image 2 Illustrates the app interface (UX) of the WMCS integrated into the map feature of the gym:

1. The legend indicates the types of weight machines (cardio machines, weights, etc.) that are in the gym through different shapes.
2. The number on the machines indicate the sets remaining by the user. This highlights the main function of the WMCS technology.
3. The WMCS technology will be utilized in gym map applications (as a feature built on top of the gym map) and the user interface will resemble the one shown on this sketch.

The mockup displayed in Image 3 illustrates a Cable Cross Machine Integrated with the WMCS feature:

1. A Cable Cross machine that is fitted with the WMCS technology.
2. The set counter fitted with the NFC chip and reader. If the user's device is not detected within a certain period of time, the user can tap their mobile devices and sync manually through the NFC tag attached to the machine.
3. The weight adjustments on the Cable Cross Machine.



Photo 1: SRSC Gym



Photo 2: SRSC Gym 2

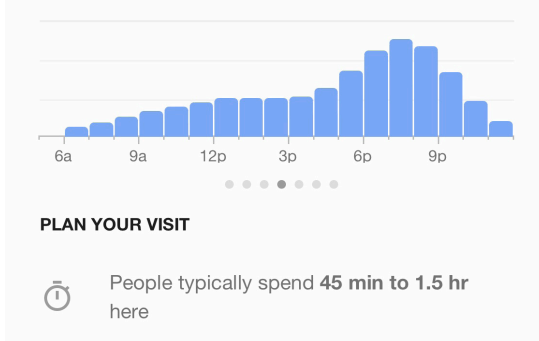


Photo 6: SRSC Traffic on Thursday

FINDINGS AND TAKEAWAYS FROM INTERVIEWS, SURVEYS AND OBSERVATIONS

Background on the interviews and surveys: The ten questions are meant to elaborate on what current gym-goers (mostly undergraduate students) think about the time they spend working out. The answers are meant to shed light upon needs and wants that are needed in current gyms (Source 1, 2, 3, 4).

Background on Observations: The observation gave me the opportunity to look at how a gym functions (particularly the weights room) on a Thursday afternoon. Throughout the observation, I was able to analyze and dissect the people on the machines and the ones that are not. The goal was to see how gym-goers interacted with the environment (Source 5).

Identifying Patterns: The biggest frustration aligns with the problems that I've outlined in the problems section of this project: certain machines are always being used. With the lack of information on the availability of equipment, gym-goers spend more time waiting for machines to open up. Yet, despite the frustration of the lack of availability, all of the respondents claimed they would not use a gym app that has the availability of the machine. This gave me the indication that even with a feature that works as seamlessly as the WMCS, certain people might not be open to the idea of it (Source 1, 2, 3, 4, 5).

The Limitations of the Interviews and Observations

All forms of primary research lacked a sufficient sample size to determine whether a feature like the WMCS would come of use. Despite all my respondents saying that they wouldn't use a gym app, with all the gym apps available today, this illustrates a point that there is a certain demographic that is willing to utilize resources that are available. During the research I've conducted, I only targeted a specific target group (college students), which led to similar answers throughout. As working out in gyms consists of varying ages, I could have obtained interviewed people from different age brackets (Source 1, 2, 3, 4, 5).

Future Applications and Current Limitations

- There are already similar implementations such as the heatmap (Source 5) and since this idea is a supplement (feature) to an already existing app, there is a degree of uncertainty of whether this feature will be utilized by others.
- Compared to Ultrawide Band and Bluetooth, NFC has a lower frequency meaning only smaller data sizes can be sent across (Source 8, 11, 13).
- Machine Huggers: They might always set the counter at a maximum, this can be remedied by setting a stricter limit to the counter.
- Similar to the WMCS, the NFC technology can be utilized to track traffic in not only a gym setting, but other venues such as theaters and retail stores (Source 8).

Works Cited

PRIMARY

Interview, Surveys, and Observations Located in Appendix 1-5:

Interview/Survey Questions (Interviews and Surveys were conducted with Individuals that workout):

1. Do you work out?
2. On average, how long are your workouts?
3. What time of the day do you go to work out?
4. How often do you work out on a weekly basis?
5. Do you have a workout plan? If not, how do you organize your workout?
6. Do you normally wait for a machine to open up or do you never encounter this problem?
7. What is your biggest frustration in a crowded gym?
8. If there is an area you wished the gym invested more in, what would it be?
9. Would you use a gym app that has the availability of the machine?
10. What's missing in your gym (anything)?

Source 1: Interview of Chaewon Lee conducted by Chae Woon Kim on 11/16/2019

Source 2: Interview of Elise Jane Lee conducted by Chae Woon Kim on 11/16/2019

Source 3: Survey of Jessy Zhang conducted by Chae Woon Kim on 11/16/2019

Source 4: Survey of Brian J. Forcier conducted by Chae Woon Kim on 11/16/2019

Source 5: Observation of Gym (SRSC) at 7-9:00pm, Thursday, Activity conducted by Chae Woon Kim on 11/07/2019

IMAGES

Image 1: Storyboard of WMCS Technology Implementation (source: Chae Woon Kim 11/17/2019)

Image 2: Gym App Interface with WMCS Feature (source: Chae Woon Kim 11/17/2019)

Image 3: Cable Cross Machine with WMCS Technology (source: Chae Woon Kim 11/17/2019)

PHOTOS

Photo 1: SRSC Gym (source: Chae Woon Kim 11/07/2019)

Photo 2: SRSC Gym 2 (source: Chae Woon Kim 11/07/2019)

Photo 3: Cable Cross Machine (source: Chae Woon Kim 11/07/2019)

Photo 4: Back Machine (source: Chae Woon Kim 11/07/2019)

Photo 5: Shoulder Press Machine (source: Chae Woon Kim 11/07/2019)

Photo 6: SRSC Traffic on Thursday (source: Chae Woon Kim 11/07/2019)

Works Cited Continued

SECONDARY

SOURCES

- Source 6: Gym Times (<https://www.usnews.com/news/articles/2013/01/03/its-gym-season-from-now-until-march>, 11/15/2019)
- Source 7: Best Time to Exercise(<https://time.com/5533388/best-time-to-exercise/>, 11/15/2019)
- Source 8: What is NFC (<https://www.techradar.com/news/what-is-nfc>, 11/15/2019)
- Source 9: Heat Map (<https://halofitness.zendesk.com/hc/en-us/articles/360010747351-Using-the-Heat-Map>, 11/15/2019)
- Source 10: Gym Membership Statista (<https://www.statista.com/statistics/236123/us-fitness-center--health-club-memberships/>, 11/15/2019)
- Source 11: Apple U1 Chip (<https://www.wired.com/story/apple-u1-chip/>, 11/15/2019)
- Source 12: NFC Enabled Mobile Devices Statista (<https://www.statista.com/statistics/461494/nfc-enabled-mobile-devices-worldwide/>, 11/15/2019)
- Source 13: What is Ultra-Wideband (<https://gizmodo.com/what-is-ultra-wideband-and-what-does-it-do-in-the-iphon-1838097808>, 11/15/2019)
- Source 14: Smart Fitness Machines (<http://nymag.com/strategist/article/smart-home-gym-equipment-mirror-tonal-peloton-review.html>, 11/15/2019)
- Source 15: Equipment Usage Tracking Technologies (<https://www.athleticbusiness.com/fitness-training/equipment-usage-tracking-technologies.html>, 11/15/2019)
- Source 16: Regular Exercise is Part of Your Job (<https://hbr.org/2014/10/regular-exercise-is-part-of-your-job>, 11/15/2019)
- Source 17: Ultrawide Band Technology (<https://www.ft.com/content/47e914a0-da3b-11e9-8f9b-77216ebe1f17>, 11/15/2019)
- Source 18: Life Fitness App (<https://lifelitness.com/home/products/lfconnect-app>, 11/15/2019)
- Source 19: Life Fitness Heat Map (<https://lifelitness.com/pressreleases/life-fitness-adds-heat-map-technology-lfconnect-platform>, 11/15/2019)
- Source 20: Average Amount of Time Spent on Sports, Exercise, and Recreation in the US from 2009 to 2018 Statista (<https://www.statista.com/statistics/189535/daily-average-time-spent-on-sports-and-exercise-in-the-us/>, 11/19/2019)
- Source 21: Attendance of Current Gym Statista (<https://www.statista.com/statistics/930495/gym-boutique-gym-membership-attendance/>, 11/17/2019)
- Source 22: NFC Enabled Cellphones (<https://www.cnet.com/news/nfc-enabled-cell-phones-to-hit-416-million-shipments-report/>, 11/17/2019)

IMAGES

- Image 4: NFC (<https://www.techradar.com/news/what-is-nfc>, 11/16/2019)
- Image 5: Rep Count App (<https://www.lifewire.com/best-workout-log-apps-4140222>, 11/17/2019)
- Image 6: Graph of NFC Enable Devices from 2012 to 2018 (<https://www.statista.com/statistics/461494/nfc-enabled-mobile-devices-worldwide/>, 11/17/2019)
- Image 7: Graph of Attendance of Current Gym (<https://www.statista.com/statistics/930495/gym-boutique-gym-membership-attendance/>, 11/17/2019)
- Image 8: Bar Graph of Total Gym Memberships in the US from 2000 to 2017 (<https://www.statista.com/statistics/236123/us-fitness-center--health-club-memberships/>, 11/17/2019)
- Image 9: Graph of Average Amount of Time Spent on Sports, Exercise, and Recreation in the US from 2009 to 2018 (<https://www.statista.com/statistics/189535/daily-average-time-spent-on-sports-and-exercise-in-the-us/>, 11/17/2019)
- Image 10: Heat Map (<https://halofitness.zendesk.com/hc/en-us/articles/360010747351-Using-the-Heat-Map>, 11/17/2019)

Appendix 1 (Source 5)

Observation 1 (Source 5)



Photo 1: SRSC Gym



Photo 2: SRSC Gym 2



Photo 3: Cable Cross Machine



Photo 4: Back Machine



Photo 5: Shoulder Press Machine

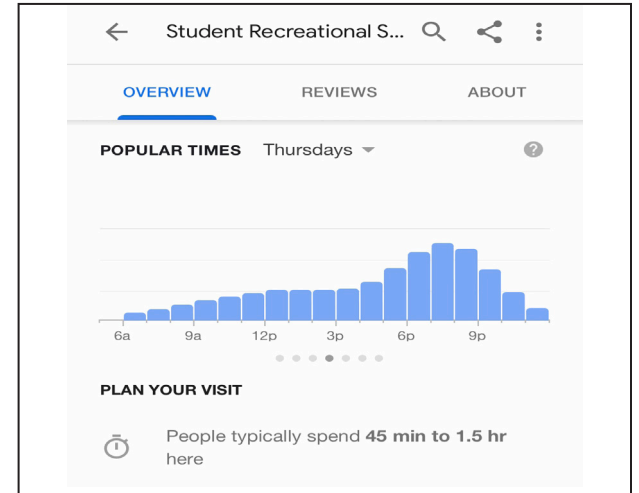


Photo 6: SRSC Traffic on Thursday

These are pictures that have aided me during my observation at the Sports Recreational Center (1st Floor Weights Room). Since the SRSC does not disclose any information regarding the how many people visit the SRSC, Image 6 was used through my phone via google search of the SRSC to determine the traffic of the location.

Appendix 1 Continued

Observation 1 Continued (Source 5)

Observation 1: The SRSC (Sports Recreational Center) 7-9pm, 1st floor of the gym (weights room), Thursday (11/07/2019)

The timeframe: 7:00 – 9:00pm, on a Thursday (11/07/2019)

The environment: Brightly lit, with most of the machines stacked side by side, with little room for anyone to walk in between.

Questions that I'm asking myself as I am conducting the observation:

- How long are individuals staying on a machine?
- How many repetitions of sets are they doing?
- What are the bystanders doing, while waiting for a machine to open up?
- How are people interacting with their environment?
- How do the gym-goers interact with the people around them?
- Are people in groups when working out?
- Are they conversing during breaks?
- Why are they standing around?

Total People within the vicinity at current time (7:00pm): 15

Classifying Individuals and How They Behaved:

Staff (2): Watching out for others that might need help while working out

Bystanders (5): Waiting for the machine to open up or with friends and waiting for their turn on the rotation.

On a Machine (8): focused on their work out and have earphones on to detain from any distractions.

Out of a possible 15 people, 8 people were solely focused on their workout and kept conversations at a minimum, once their session with the machine was over, they would move on to the next equipment. Every action that they conducted during my observation indicated that they wanted to maximize every minute of their time at the gym. The only time they would stop was to go grab water or check the time or song track on their phones.

The two staff members were on standby to check if the rules were being followed the gym-goers. During my time nobody was in need of any help.

The people that were “Bystanders” would either stand next to the machine and wait, or roam around the vicinity till the equipment became available. They would often ask how many sets the person on the machine had but would not leave the vicinity. They would check their phones constantly and would often times seem impatient and move onto another machine by the time the person finished their session on the machine. Sometimes a bystander would be with other friends and would converse with them while waiting.

From the observation, it is apparent that the availability of the machines had a large role in whether an individual would be focused or standing around. During the time I was conducting the interview, the machines were available, but even with machines open, certain equipment had a cue of people waiting.

Appendix 2 (Source 1)

Interview Result 1

Chaewon Lee, Senior at Indiana University

1. Do you work out?

Yes, I want to stay fit.

2. How long are your workouts?

Roughly 2 hours

3. What time of the day do you go to workout?

In the morning or between 3-5pm or when I have a chunk of time available during the day.

4. How often do you workout on a weekly basis?

5-6 times a week

5. Do you have a workout plan? If not, how do you organize your workout?

Yes, I follow a plan from my trainer.

6. Do you normally wait for a machine to open up or do you never encounter this problem?

Usually in the afternoons, I have to wait for a machine to open up, but not for a long time.

7. What is your biggest frustration in a crowded gym?

Space and machines/equipment always being used. Some people like to stay on one machine for a particular amount of time that could annoy others.

8. If there is an area you wished the gym invested more in, what would it be?

More gym equipment other than machines like resistance bands, trx bands, sleds, etc.

9. Would you use a gym app that has the availability of the machine?

Probably not. The gym isn't that big, so I could just look and availability is not a huge concern.

10. What's missing in your gym (anything)?

Space and equipment for circuit training.

Appendix 3 (Source 2)

Interview Result 2

Elise Jane Lee, Junior at Indiana University

1. Do you work out?

Yes

2. How long are your workouts?

I typically work out anywhere from 45 minutes-1 hour depending on how much time I have.

3. What time of the day do you go to workout?

Since my classes are later in the day, I work out early in the morning around 8 am.

4. How often do you workout on a weekly basis?

On a slow week I'll be able to work out 4 times a week, but on a busy week I'll only workout a couple days out of the week.

5. Do you have a workout plan? If not, how do you organize your workout?

Yes

6. Do you normally wait for a machine to open up or do you never encounter this problem?

If a machine is busy I'll wait and do another part of my workout before going back and checking if the machine is open.

7. What is your biggest frustration in a crowded gym?

My biggest frustration is when all the machines are taken and it takes longer to complete my workout because of this. Sometimes, I don't even get to finish my workout and have to leave, because of time.

8. If there is an area you wished the gym invested more in, what would it be?

I wish the gym would invest more in their weights section as it is a smaller portion of the gym that a lot of people use.

9. Would you use a gym app that has the availability of the machine?

I probably would use an app that let me know if machinery is open as I think that would save time and make workouts more efficient. However, I don't see myself religiously using it everytime I go to the gym.

10. What's missing in your gym (anything)?

I don't feel that anything is missing in my gym.

Appendix 4 (Source 3)

Survey Result 1

Jessy Zhang, Sophomore at Indiana University

1. Do you work out?

Yes

2. How long are your workouts?

1-2 hours

3. What time of the day do you go to workout?

6am morning times

4. How often do you workout on a weekly basis?

6 times a week

5. Do you have a workout plan? If not, how do you organize your workout?

Yes, each day I work on a certain muscle category

6. Do you normally wait for a machine to open up or do you never encounter this problem?

I wait for the machines but this rarely happens

7. What is your biggest frustration in a crowded gym?

That people are staring and too close to my personal space

8. If there is an area you wished the gym invested more in?

More stretching machines

9. Would you use a gym app that has the availability of the machine?

No because I try not to be on my phone when I'm at the gym because of distractions

10. What's missing in your gym (anything)?

Boxing bag, jump rope

Appendix 5 (Source 4)

Survey Result 2

Brian J. Forcier, Senior at Indiana University

1. Do you work out?

Yes

2. How long are your workouts?

1 hour

3. What time of the day do you go to workout?

6pm

4. How often do you workout on a weekly basis?

5 times a week

5. Do you have a workout plan? If not, how do you organize your workout?

Yes, muscle for life workouts

6. Do you normally wait for a machine to open up or do you never encounter this problem?

No, usually can find one or ask to work in

7. What is your biggest frustration in a crowded gym?

Sometimes can't use machines

8. If there is an area you wished the gym invested more in?

Invest in more free weight equipment like squat decks and deadlift spots

9. Would you use a gym app that has the availability of the machine?

No, an app would not be feasible

10. What's missing in your gym (anything)?

Pretty satisfied